Jhezra A. Tolentino

Software Engineering | AI Engineering

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EDUCATION

FEU Institute of Technology — BS Computer Science, Specialization in Software Engineering *2023 – Expected 2027*

- GPA 3.8 / 4.00 · Consistent Gold Dean's Lister · University Elite Scholar · DOST-SEI Scholar · Top Performing Student: Rank 2 (A.Y. 2023–2024), Rank 8 (2024–2025), Rank 3 (2025–2026)

EXPERIENCE

Software Engineering Intern, Ed3N Ventures — Onsite *Jul 2026 – Present*

Creator Loans (pre-launch): a lending marketplace that turns a TikTok Shop creator's sales into a verified income figure and a borrowing limit, then lets licensed lenders bid on the loan. NestJS, React, TypeScript, PostgreSQL.

- Designed income scoring from TikTok Shop sales and commission rate. Required a second source (bank statement or payout record) before an admin can list a creator, to strengthen the creator's credibility.
- Designed blind lender bidding: a creator's identity stays hidden until the creator accepts an offer. Made the API grade every dashboard figure as measured, derived, or assumed, with tests that fail on an ungraded figure.
- Built the creator, admin, and lender apps and the API in four weeks (Sep – Oct 2026). Authored 184 of 292 merged pull requests, wrote about 90% of the 45 architecture decision records, and kept about 2,600 automated tests as the merge gate.

onnit.ai (Agent Provisioning Platform): a personal AI agent product. Backend and agent onboarding.

- Reverse-engineered the API of an undocumented third-party agent platform, including its session-token auth scheme and multi-stage pairing lifecycle. This enabled real Telegram and WhatsApp QR onboarding with no frontend changes.
- Created the onboarding system prompt for the agent that onboards new users.

Lead Web Developer, FEU Tech ACM Student Chapter *Aug 2025 – Aug 2026*

- Led a team of beginner and junior developers on architecture and skill building. Built a Next.js 14 and TypeScript e-commerce platform for official ACM merchandise, used by 400+ members.

PROJECTS

KitaKo (Undergraduate Thesis) — Team Lead and ML/Software Engineer *Dec 2025 – May 2026*

An offline Android photo-search app that understands Taglish (Tagalog-English) queries.

- Led the thesis team and designed the research methodology. Collected images for the 109,789-image Taglish dataset used to fine-tune the model, with the team's custom scraper on publicly licensed sources.
- Built the on-device search layer: HNSW (C++ hnswlib through Dart FFI) and IVF-PQ indexing. At equal Recall@1, HNSW cut search time from 28.34 ms to 6.67 ms. IVF-PQ cut the 10,000-image index size by 91.45%.
- Integrated the team's fine-tuned SigLIP-2 model into the Flutter app with ONNX Runtime. On a 300-query human-written benchmark, it beat SigLIP and SigLIP-2 on all 7 metrics (Recall@1 0.41 vs 0.27; mAP 0.78 vs 0.63).
- Team results: Taglish and English results became statistically the same (mAP $p = .920$, $d = -0.010$), and the unevenness across languages (TVD@1) fell 73.5%. Usability scored 90.95 (SUS, 55 users).

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, C++, Dart, SQL | **Frameworks & Libraries:** NestJS, Next.js, React, FastAPI, Flutter, Tailwind CSS, Drizzle ORM
- **AI / ML:** PyTorch, SigLIP-2, ONNX Runtime, HNSW / IVF-PQ ANN indexing | **Tools:** PostgreSQL, Neon, Git, pnpm, Turborepo, Vitest, Playwright
- **AI-Assisted Development:** Claude Code (agent workflows, code review, test-gated changes)

AWARDS & HONORS

- Champion — CyberXploit, FEU Tech *2026*
- 1st Runner-Up — GDG PUP SEEN, Capture-the-Flag *2026*
- 3rd Place — FEU Tech Open AI Campus Hackathon *2026*
- TOPCIT Level 3 (Competent) — 13th National Examination *2025*
- Finalist — Algolympics, National Competitive Programming *2025*
- Finalist — Start-A-Ton, START-DOST Datathon *2026*
- 3rd Runner-Up — First Gen Code Green, National Pitching *2024*

LEADERSHIP & CERTIFICATIONS

- **Director for Academics, FEU Tech ACM Student Chapter** (Aug 2024 – Aug 2025): led the academics committee. Ran review sessions, peer tutorials, and institute competitions with several committees.
- **Certifications:** DataCamp AI Engineer for Data Scientists · DataCamp Associate Data Analyst · IT Specialist: Python (Certiport) · PMI Project Management Ready